

AutoDrip: A Low-Cost Dengue Fluid Management System

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Abstract. Dengue fever is a major health concern in subtropical regions, where severe cases require precise intravenous (IV) fluid therapy to prevent life-threatening complications. In resource-limited settings, manual drip regulation is error-prone due to staff shortages and inadequate monitoring. AutoDrip, an automated gravity-driven fluid management system, is presented to calculate the required flow rate (RFR) and regulate the clinical flow rate (CFR) for dengue patients. The device combines a stepper motor-driven flow controller, a non-contact infrared (IR) drop sensor, and a binary search-based feedback algorithm. System performance was evaluated under dengue IV therapy protocols. The RFR calculation achieved 100% accuracy, drop count accuracy averaged 98.3%, and CFR regulation reached 99.3% accuracy, stabilizing within 7-9 iterations. These results demonstrate that precise and reliable infusion control can be achieved with a low-cost device, making AutoDrip a practical solution to enhance patient safety and reduce clinical workload during dengue outbreaks.

KEY WORDS: Automated IV therapy, Dengue fluid management, Medical device automation, Healthcare resource optimization, Point-of-care technology

1. INTRODUCTION

Dengue is a major public health concern in tropical regions such as Bangladesh, with 321,073 cases recorded in 2023—the country's largest outbreak to date [1]. Severe cases require precise intravenous (IV) fluid therapy, yet manual regulation with gravity sets is highly error-prone, often deviating by ± 20 –25%, particularly during patient surges that strain nursing capacity. Infusion pumps provide accurate delivery but remain costly and impractical in resource-limited settings.

Several automated infusion systems have been proposed, using optical, ultrasonic, or pressure sensors with servo or peristaltic pump mechanisms [2–4]. Al-Obaidi et al. (2024) developed an Intelligent Saline Controlling Valve utilizing a Mamdani fuzzy inference system to autonomously regulate intravenous fluid rates and prevent air embolisms by monitoring drip speed, pressure, and bubble presence [5]. Pabitha and Selvakumar (2025) proposed an IoT-enabled architecture facilitating real-time remote monitoring of IV fluid levels and automated alerting via a mobile application [6]. Gravity-based solutions such as *AccuDrip* demonstrate improved stability over manual regulation [7]. However few designs specifically target dengue requirements or include formal tolerance and convergence analysis. To address this gap, we propose **AutoDrip**, a low-cost, gravity-driven automated controller. The system integrates (i) a flow calculator based on national dengue guidelines, (ii) a non-contact infrared drop sensor, (iii) a stepper-actuated flow regulator, and (iv) a binary search-based feedback algorithm. AutoDrip is designed to ensure accurate, safe, and standardized fluid delivery for Group B dengue patients in resource-constrained healthcare environments.

2. METHODOLOGY

2.1 FLOW RATE CALCULATION

2.1.1 Clinical Classification of Dengue Patients

Dengue, an arboviral disease, is transmitted by Aedes mosquitoes. Patients are stratified as Group A (mild), Group B (warning signs or risk factors requiring hospitalization), and Group C (severe, needing intensive care) [8,9]. This system targets Group B dengue patients, who require guided IV fluid therapy based on national clinical guidelines [8-10].

2.1.2 Calculation Principle

For Group B dengue patients, fluid requirement is calculated from demographic and clinical inputs (age, sex, height, body weight, blood pressure, capillary refill time, and IV set drip factor). Eligibility is confirmed when **pulse pressure >20 mmHg** and **capillary refill time <2 s** [1]. Maintenance fluid is then determined using the **Holliday–Segar method**, with actual body weight applied when BMI ≤ 25 , otherwise **IBW** (Ideal Body Weight) is derived from the formula of equation (1) and (2),

$$\text{Men: } 50 + \max(0, 0.91 \times [\text{Height (cm)} - 152.4]) \text{ kg} \quad (1)$$

$$\text{Women: } 45.5 + \max(0, 0.91 \times [\text{Height (cm)} - 152.4]) \text{ kg} \quad (2)$$

Maintenance and Deficit Fluid Requirement

The **Holliday–Segar 4-2-1 rule** was used: 4 mL·kg⁻¹·h⁻¹ for the first 10 kg, 2 mL·kg⁻¹·h⁻¹ for the next 10 kg, and 1 mL·kg⁻¹·h⁻¹ thereafter. Deficit Fluid Requirement (Oral+IV) for ‘Group-B NS’ patients, a default 5% dehydration is assumed in equation (3):

$$\text{Deficit fluid (mL)} = 50 \text{ mL} \cdot \text{kg}^{-1} \times W_{t_{\text{param}}}(\text{kg}) \quad (3)$$

Flow rate prescriptions follow the national dengue guideline, with reassessment every 6 hours for at least 48 hours. [9]

2.2 FLOWRATE COUNTER

The AutoDrip device monitors IV drip rates using a non-contact infrared (IR) sensor composed of an LED emitter and photodiode receiver. With the fall of each droplet, the LED is interrupted, generating a signal for drop counting. Since drop rates are uniform, continuous long-term monitoring is unnecessary. Tests with different averaging windows (5–20 s) showed that **10 s (T_{avg})** provides stable and accurate measurement. Drop counts are recorded in drops/s and converted to drops/min for comparison with prescribed rates. This method avoids contamination and integrates seamlessly with standard IV sets without modification.

2.3 FLOW CONTROL MECHANISM

The core of the AutoDrip system is its automated flow rate control mechanism, which translates digital calculations into precise mechanical adjustments of the intravenous (IV) fluid flow.

The primary components of the Flow Control Mechanism are:

- **Stepper Motor(101):** The final AutoDrip device employs a Nema 17 stepper motor (17HS4401S) with 600-step precision for flow control. The stepper, coupled with a custom conical coupler and dial flow meter, ensures reliable and accurate performance.
- **Dial Flow Meter (114):** The system uses a medically approved rotational dial flow meter to regulate IV tube constriction. While its built-in scale is not used for setting flow rates, the device ensures safe and reliable regulation as part of the AutoDrip system.

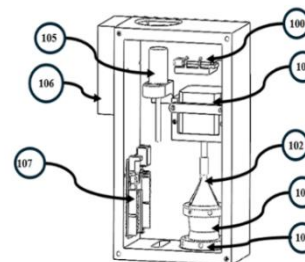


FIGURE 1 Isometric View of AutoDrip Device Assembly

- **Conical Coupler (113):** A custom conical coupler connects the stepper motor to the dial flow meter, ensuring balanced torque transfer with minimal friction.
- **Fixer (115):** To ensure stability and prevent misalignment during rotation, a fixer is used to immobilise the stationary lower part of the dial flow meter assembly within the device housing.

2.4 FEEDBACK AND CONTROL ALGORITHM

2.4.1 Problem setup

Let $\theta \in [\theta_{min}, \theta_{max}]$ denote the motor angle controlling the tube constriction, and let $f(\theta)$ represent the steady-state drop rate measured by the IR sensor (drops/min). The clinical target is f_t . Within the operating region of interest, the mechanism behaves **monotonically**; that is, if $\theta_1 < \theta_2$, then $f(\theta_1) \geq f(\theta_2)$. The objective is to determine an angle θ_t such that $|f(\theta_t) - f_t| \leq \delta$, where δ is a tolerance derived from clinical acceptance. With stepper-driven actuation, the admissible set of angles is shown in equation (4):

$$\theta = \{\theta_{min} + kc \mid k = 0, 1, 2, \dots, N-1\}, N = \left\lceil \frac{\theta_{max} - \theta_{min}}{\Delta\theta} \right\rceil \quad (4)$$

2.4.2 Binary search controller

Define the signed error $e(\theta) = f(\theta) - f_t$. With monotonic $f(\theta)$, the domain partitions into regions above or below the target. The binary search algorithm proceeds on the index interval $[L, R] \subseteq \{0, 1, 2, \dots, N-1\}$:

1. Initialize $L = 0, R = N - 1$
2. While $L \leq R$:

- $M = \left\lfloor \frac{L+R}{2} \right\rfloor \rightarrow$ set $\theta = \theta_{min} + M\Delta\theta$.
- Measure $f(\theta)$. (Averaged over a window T_{avg})
- If $|f(\theta_t) - f_t| \leq \delta \rightarrow$ return θ
- If $f(\theta) > f_t$: increase constriction $\rightarrow$ set $L = M + 1$
- Else: decrease constriction $\rightarrow$ set $R = M - 1$

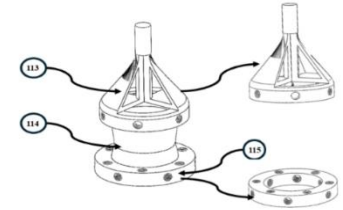


FIGURE 2 Controller-Coupler Mechanism

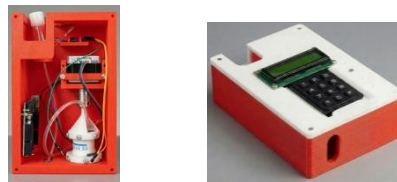


FIGURE 3 Internal view (left) and external view (right) of the assembled device

2.4.3 Stabilisation Time

Each evaluation consists of, $T_{eval} = T_{move} + T_{settle} + T_{avg}$. Each iteration of the algorithm halves the interval, keeping number of measurements K bounded by, $\lceil \log_2 N \rceil$. Since averaging dominates ($T_{avg} \approx 10s$ window to smooth drop-count noise), the stabilization time, T_{bin} is bounded by, $\lceil \log_2 N \rceil \cdot T_{eval}$.

Illustration for a 300° dial range with $\Delta\theta = 1^\circ$ where, $N \approx 300, K \leq \lceil \log_2 300 \rceil, T_{eval} \approx 10s$

$$T_{bin} \lesssim 9 * 10 = 90s \quad (5)$$

2.4.4 Step-Size Adequacy and Angle-Flow Relation

The binary search algorithm operates in $\Delta\theta = 1^\circ$ increments, requiring verification that such steps remain within clinical tolerance. Let $Q(\theta)$ be the flow rate ($mL \cdot h^{-1}$) and the step size can be determined from equation (6):

$$S_{1^\circ} = |Q(\theta + 1^\circ) - Q(\theta)|, \quad S_{max} = \max_{\theta} S_{1^\circ}(\theta) \quad (6)$$

The step size safety condition is therefore,

$$S_{max} \cdot \Delta\theta \leq \tau Q_t, \quad \tau = 0.05(\pm 5\%) \quad (7)$$

With Q_t being the target flow-rate consistent with infusion pump standards. From the dial curve in fig. 4, $S_{max} \approx 5 mL \cdot h^{-1}$ near $225\text{--}250 mL \cdot h^{-1}$. Table 2 confirms that increments across $50\text{--}250 mL \cdot h^{-1}$ remain well within $\pm 5\%$, with deviations only $0.5\text{--}2.1\%$ per degree.

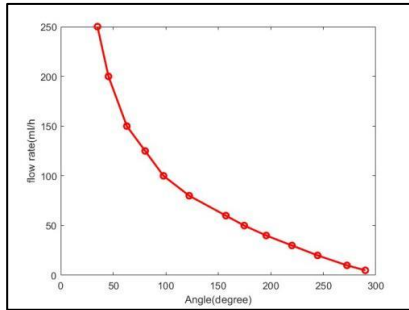


FIGURE 4: Flow-rate vs angle curve of a standard Dial-a-Flow regulator.

TABLE 2: Safety check for 1° steps across target flows

Target Flow Q_t (mL/h)	$\pm 5\%$ Tolerance Band (mL/h)	Observed ΔQ per 1° (mL/h)
50	± 2.5	0.5
100	± 5	1.0
200	± 10	2.5
250	± 12.5	5.0

A fitted power law model can be given like equation (8),

$$Q(\theta) = a(\theta - \theta_0)^{-b}, \quad a = 6.24 \times 10^9, b = 3.22, \theta_0 = -163.5^\circ, R^2 = 0.992 \quad (8)$$

Further supports this. For 1° step, the relative change per degree simplifies to equation (9),

$$\frac{1}{Q} \left| \frac{dQ}{d\theta} \right| = \frac{b}{\theta - \theta_0} \quad (9)$$

predicting relative slopes of $0.7\text{--}1.6\%$ per degree over the measured range. Both empirical and modeled results confirm that the 1° step resolution is clinically safe.

2.4.5 Effect of Height (hydrostatic head)

For gravity infusion, height scales the entire angle-flow curve multiplicatively following the eq (10).

$$Q(\theta, h) = K(h) \cdot a(\theta - \theta_0)^{-b}, \quad K(h) \propto (h - h_v)^\alpha, \alpha \in [0.5, 1] \quad (10)$$

where h is the fluid level above the vein, and h_v is the effective venous pressure height. The relative slope can be determined from equation (11)

$$\frac{1}{Q} \left| \frac{dQ}{d\theta} \right| = \frac{K(h) \cdot ab(\theta - \theta_0)^{-b-1}}{K(h) \cdot a(\theta - \theta_0)^{-b}} = \frac{b}{\theta - \theta_0} \quad (11)$$

showing that the **percentage change per degree is independent of height**. Therefore, the 0.71–1.62% per-degree range holds for all clinically relevant heights.

3. RESULTS

3.1 Flow Rate Detection Accuracy

The IR-based drop counter was validated against manual observation at four flow settings (10, 20, 30, and 50 drops/min). Figure 5 shows an almost perfect linear correlation between sensor output and manual counts ($R^2 \approx 0.998$), confirming reliable drop detection. Deviations were minimal, with IR readings differing from manual counts by at most ± 2 drops/min. This demonstrates that the non-contact optical sensor provides accurate and sterile monitoring of IV flow rates in the clinically relevant range.

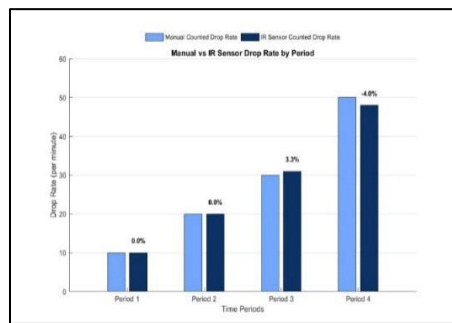


FIGURE 5: Manual vs IR sensor drop rate detection by period

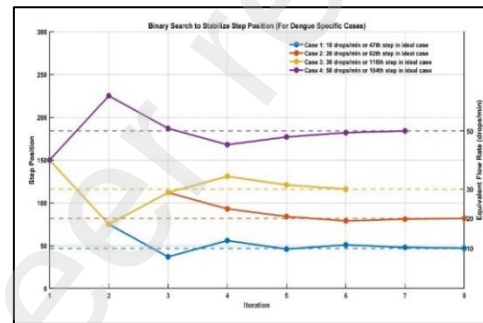


FIGURE 6: Binary Search to stabilize Step position (For dengue specific cases)

3.2 Feedback Loop Convergence

The closed-loop binary search controller consistently stabilized the infusion flow to within $\pm 5\%$ of the prescribed rate. Convergence occurred within 5–9 iterations, corresponding to a worst-case stabilization time of 90s (Figure 6). This aligns with theoretical predictions in equation (5) and demonstrates the efficiency of binary search compared to sequential or proportional methods. The evaluation was carried out under ideal-case conditions for dengue-specific flow rates, ensuring that the results directly reflect the clinical use scenario.

3.3 Step-Size Tolerance Analysis

Analysis of the angle–flow curve showed that the steepest region (225–250 mL/h) produced a maximum step sensitivity of ~ 5 mL/h per 1° , corresponding to $\sim 2.1\%$ of the flow at that setting. Across the full 50–250 mL/h range, relative deviations per degree remained between 0.7% and 2.1%, well within the $\pm 5\%$ tolerance band. Thus, the 1° actuator resolution is adequate for all clinically relevant infusion rates.

4. DISCUSSION

The closed-loop controller reliably converged within $\pm 5\%$ of prescribed dengue-specific infusion rates (10–50 drops/min, equivalent to 40–200 mL/h), stabilising in fewer than nine iterations (~ 90 s). Even under an edge case

of ~4% flow count error around 180–200° on the dial, the relative sensitivity was only ~0.9% per degree—well below the tolerance band. At the upper end (200 mL/h, 50 drops/min), the steepest step sensitivity observed was ~2.1% per degree, which also remains within safety limits. Together with ± 2 drops/min sensor accuracy, these results confirm that the system maintains safe operation across the entire dengue treatment range.

Clinically, this performance is significant as manual gravity regulators deviate by ± 20 –25%, whereas AutoDrip achieves pump-grade accuracy using only low-cost hardware. The ~90 s stabilization period derived in equation (9) is negligible in practice, as dengue infusion protocols reassess and modify flow rates at 6-hour intervals [9]; therefore, a 1.5 minute convergence has no meaningful impact on total fluid administration. Although bag height influences the hydrostatic head, the controller adaptively compensates for these changes by adjusting constriction to maintain the prescribed flow rate. Consideration of venous back-pressure effects was beyond the scope of this study.

5. CONCLUSION

In conclusion, this project successfully demonstrates AutoDrip as a cost-efficient automated infusion system that addresses the critical challenge of precise fluid management for dengue patients, particularly in resource-constrained settings. By integrating non-contact IR sensing for real-time drop detection with a stepper motor-driven feedback controller, the system achieves remarkable accuracy and reliability. The results confirm the system's clinical significance. The feedback loop consistently converges to within $\pm 5\%$ of the prescribed rate, while the IR sensor maintains a minimal deviation of at most ± 2 drops/min. This performance represents a transformative improvement over manual gravity regulators, which commonly deviate by a hazardous ± 20 –25%. Furthermore, a step-size tolerance analysis revealed that the system's mechanical sensitivity remains between **0.7% and 2.1%** per degree, ensuring stability and safety even at steeper flow regions. These outcomes collectively emphasize the critical role of accessible automation in elevating IV fluid management to a reliable, pump-grade process without the high cost. Ultimately, integrating this technology into clinical practice has the potential to alleviate critical personnel shortages, enhance the standard of care, and provide a tailored, scalable solution to mitigate the burden of dengue epidemics.

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